

Q1 Language of kinematics

Name: _____

Class: _____

Date: _____

Time: **423 minutes**

Marks: **352 marks**

Comments:

EXAM PAPERS PRACTICE

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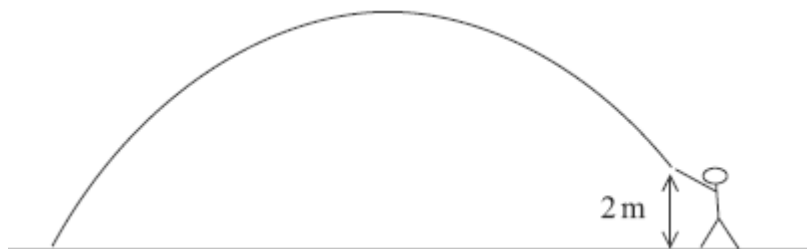
Q1.

A cricket ball is hit at ground level on a horizontal surface. It initially moves at 28 m s^{-1} at an angle of 50° above the horizontal.

- (a) Find the maximum height of the ball during its flight.

(4)

- (b) The ball is caught when it is at a height of 2 metres above ground level, as shown in the diagram.



Show that the time that it takes for the ball to travel from the point where it was hit to the point where it was caught is 4.28 seconds, correct to three significant figures.

(5)

- (c) Find the speed of the ball when it is caught.

(5)

(Total 14 marks)

Q2.

A particle moves along a straight line. At time t , it has velocity v , where

$$v = 4t^3 - 8 \sin 2t + 5$$

When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t .

(Total 4 marks)

Q3.

A particle moves on a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

At time t seconds, the position vector of the particle is $\mathbf{r}$ metres, where

$$\mathbf{r} = \left(2e^{\frac{1}{2}t} - 8t + 5 \right) \mathbf{i} + (t^2 - 6t) \mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t .

(3)

- (b) (i) Find the speed of the particle when $t = 3$. (2)
- (ii) State the direction in which the particle is travelling when $t = 3$. (1)
- (c) Find the acceleration of the particle when $t = 3$. (3)
- (d) The mass of the particle is 7 kg.

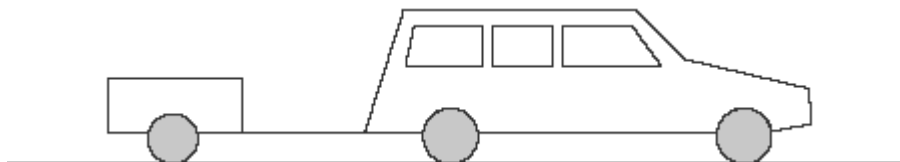
Find the magnitude of the resultant force on the particle when $t = 3$.

(3)

(Total 12 marks)

Q4.

A car, of mass 1400 kg, is towing a trailer, of mass 600 kg. The two vehicles accelerate together at 1.3 m s^{-2} along a straight horizontal road.

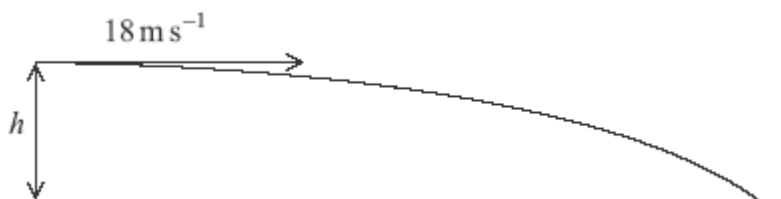


- (a) Find the distance that the car and trailer would travel while accelerating from rest to 13 m s^{-1} . (3)
- (b) A forward driving force, of magnitude 3900 N, acts on the car. A resistance force, of magnitude 800 N, also acts on the car.
- (i) A resistance force, of magnitude P newtons, acts on the trailer. Find P . (3)
- (ii) Find the magnitude of the force that the car exerts on the trailer. (3)

(Total 9 marks)

Q5.

An arrow is fired horizontally at a speed of 18 m s^{-1} from a point at a height of h metres above horizontal ground. The arrow hits the ground after it has been moving for 0.6 seconds. Model the arrow as a particle that moves only under the influence of gravity.



- (a) Show that $h = 1.76$, correct to three significant figures. (2)
- (b) Find the horizontal distance travelled by the arrow during its flight. (2)
- (c) Find the speed of the arrow and the direction in which it is moving when it hits the ground. (6)
- (Total 10 marks)

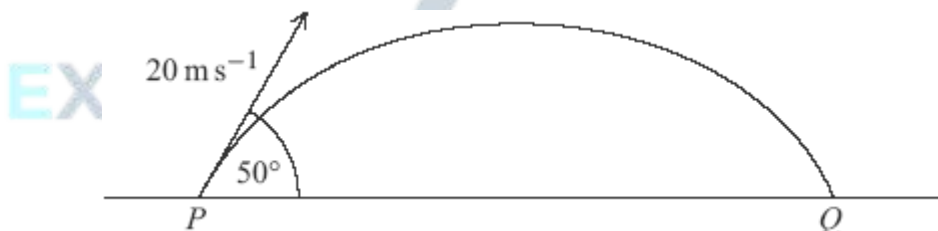
Q6.

A lift is travelling upwards and accelerating uniformly. During a 5 second period, it travels 16 metres and the speed of the lift increases from $u \text{ m s}^{-1}$ to 4.2 m s^{-1} .

- (a) Find u . (3)
- (b) Find the acceleration of the lift. (3)
- (Total 6 marks)

Q7.

A ball is kicked from the point P on a horizontal surface. It leaves the surface with a velocity of 20 m s^{-1} at an angle of 50° above the horizontal and hits the surface for the first time at the point Q . Assume that the ball is a particle that moves only under the influence of gravity.



- (a) Show that the time that it takes the ball to travel from P to Q is 3.13 s, correct to three significant figures. (4)
- (b) Find the distance between the points P and Q . (2)
- (c) If a heavier ball were projected from P with the same velocity, how would the distance between P and Q , calculated using the same modelling assumptions, compare with your answer to part (b)? Give a reason for your answer. (2)
- (d) Find the maximum height of the ball above the horizontal surface.

(3)

- (e) State the magnitude and direction of the velocity of the ball as it hits the surface.

(2)

(Total 13 marks)

Q8.

A pellet is fired from a window at a height of 3 metres above horizontal ground. Initially, the pellet travels at 70 m s^{-1} at an angle of 10° above the horizontal.

- (a) Show that the time for which the pellet travels before it hits the ground is 2.71 seconds, correct to three significant figures.

(6)

- (b) Find the horizontal distance that the pellet travels before it hits the ground.

(2)

- (c) Find the minimum speed of the pellet during its flight.

(2)

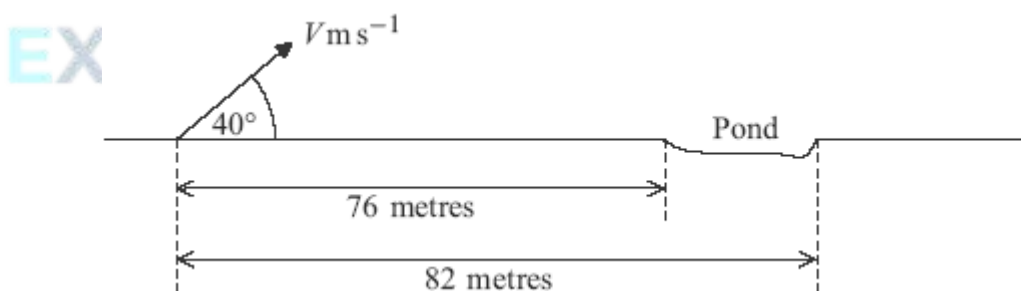
- (d) Find the speed of the pellet when it hits the ground.

(4)

(Total 14 marks)

Q9.

A golfer hits a ball which is on horizontal ground. The ball initially moves with speed $V \text{ m s}^{-1}$ at an angle of 40° above the horizontal. There is a pond further along the horizontal ground. The diagram below shows the initial position of the ball and the position of the pond.



- (a) State **two** assumptions that you should make in order to model the motion of the ball.

(2)

- (b) Show that the horizontal distance, in metres, travelled by the ball when it returns to ground level is

$$\frac{V^2 \sin 40^\circ \cos 40^\circ}{4.9}$$

(6)

- (c) Find the range of values of V for which the ball lands in the pond.

(4)

(Total 12 marks)

Q10.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. A helicopter moves horizontally with a constant acceleration of $(-0.4\mathbf{i} + 0.5\mathbf{j}) \text{ m s}^{-2}$. At time $t = 0$, the helicopter is at the origin and has velocity $20\mathbf{i} \text{ m s}^{-1}$.

- (a) Write down an expression for the velocity of the helicopter at time t seconds.

(2)

- (b) Find the time when the helicopter is travelling due north.

(3)

- (c) Find an expression for the position vector of the helicopter at time t seconds.

(2)

- (d) When $t = 100$:

- (i) show that the helicopter is due north of the origin;

(3)

- (ii) find the speed of the helicopter.

(3)

(Total 13 marks)

Q11.

A cricket ball is hit from the floor of a sports hall, which has a height of 6 metres. The initial velocity of the ball is 20 m s^{-1} at an angle of 60° above the horizontal.

Assume that the cricket ball is a particle which moves only under the influence of gravity.

- (a) Show that the ball hits the ceiling of the sports hall approximately 0.389 seconds after it was hit.

(5)

- (b) Find the horizontal distance travelled by the ball before it hits the ceiling.

(2)

- (c) Find the speed of the ball just before it hits the ceiling.

(5)

(Total 12 marks)

Q12.

A lift rises vertically from rest with a constant acceleration.

After 4 seconds, it is moving upwards with a velocity of 2 m s^{-1} .

It then moves with a constant velocity for 5 seconds.

The lift then slows down uniformly, coming to rest after it has been moving for a total of 12 seconds.

- (a) Sketch a velocity–time graph for the motion of the lift.

(4)

- (b) Calculate the total distance travelled by the lift.

(2)

- (c) The lift is raised by a single vertical cable. The mass of the lift is 300 kg. Find the maximum tension in the cable during this motion.

(4)

(Total 10 marks)

Q13.

A golf ball is struck from a point on horizontal ground so that it has an initial velocity of 50 m s^{-1} at an angle of 40° above the horizontal.

Assume that the golf ball is a particle and its weight is the only force that acts on it once it is moving.

- (a) Find the maximum height of the golf ball.

(4)

- (b) After it has reached its maximum height, the golf ball descends but hits a tree at a point which is at a height of 6 metres above ground level.



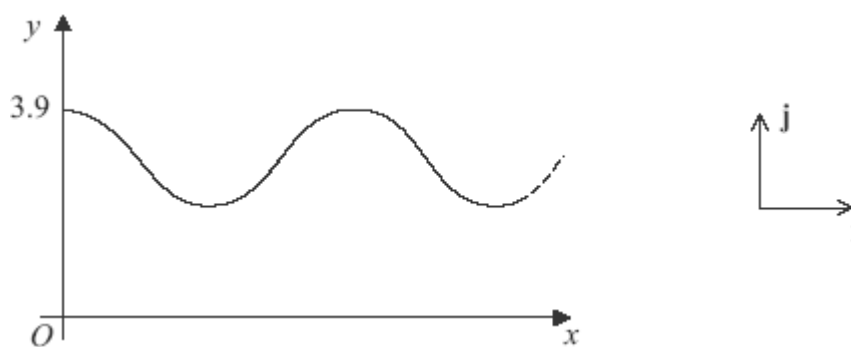
Find the time that it takes for the ball to travel from the point where it was struck to the tree.

(6)

(Total 10 marks)

Q14.

Jane is on a ride in a theme park. Part of the curved path of the ride is shown in the diagram.



Jane's position vector, $\mathbf{r}$ metres, at time t seconds is given by

$$\mathbf{r} = 1.2t \mathbf{i} + (3 + 0.9 \cos t) \mathbf{j}$$

where the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed horizontally and vertically upwards respectively.

- (a) Find an expression for Jane's velocity at time t . (2)
- (b) (i) Find an expression for Jane's speed at time t . (2)
- (ii) Find Jane's maximum speed during the ride. (2)
- (Total 6 marks)

Q15.

An arrow is fired from a point A with a velocity of 25 m s^{-1} , at an angle of 40° above the horizontal. The arrow hits a target at the point B which is at the same level as the point A , as shown in the diagram.



- (a) State **two** assumptions that you should make in order to model the motion of the arrow. (2)
- (b) Show that the time that it takes for the arrow to travel from A to B is 3.28 seconds, correct to three significant figures. (4)
- (c) Find the distance between the points A and B . (2)
- (d) State the magnitude and direction of the velocity of the arrow when it hits the target. (2)

- (e) Find the minimum speed of the arrow during its flight.

(2)

(Total 12 marks)

Q16.

A boat is initially at the origin, heading due east at 5 m s^{-1} . It then experiences a constant acceleration of $(-0.2\mathbf{i} + 0.25\mathbf{j}) \text{ m s}^{-2}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) State the initial velocity of the boat as a vector.

(1)

- (b) Find an expression for the velocity of the boat t seconds after it has started to accelerate.

(2)

- (c) Find the value of t when the boat is travelling due north.

(3)

- (d) Find the bearing of the boat from the origin when the boat is travelling due north.

(6)

(Total 12 marks)

Q17.

A golf ball is projected from a point O with initial velocity V at an angle α to the horizontal. The ball first hits the ground at a point which is at the same horizontal level as O , as shown in the diagram.



It is given that $V \cos \alpha = 6u$ and $V \sin \alpha = 2.5u$.

- (a) Show that the time taken for the ball to travel from O to A is $\frac{5u}{g}$.

(4)

- (b) Find, in terms of g and u , the distance OA .

(2)

- (c) Find V in terms of u .

(2)

- (d) State, in terms of u , the least speed of the ball during its flight from O to A .

(1)

(Total 9 marks)

Q18.

A particle P moves with acceleration $(-3\mathbf{i} + 12\mathbf{j}) \text{ m s}^{-2}$. Initially the velocity of P is $4\mathbf{i} \text{ m s}^{-1}$.

- (a) Find the velocity of P at time t seconds.

(2)

- (b) Find the speed of P when $t = 0.5$.

(3)

(Total 5 marks)

Q19.

A car travels along a straight horizontal road. The motion of the car can be modelled as three separate stages.

During the first stage, the car accelerates uniformly from rest to a velocity of 10 m s^{-1} in 6 seconds.

During the second stage, the car travels with a constant velocity of 10 m s^{-1} for a further 4 seconds.

During the third stage of the motion, the car travels with a uniform retardation of magnitude 0.8 m s^{-2} until it comes to rest.

- (a) Show that the time taken for the **third** stage of the motion is 12.5 seconds.

(2)

- (b) Sketch a velocity–time graph for the car during the three stages of the motion.

(4)

- (c) Find the total distance travelled by the car during the motion.

(3)

(Total 9 marks)

Q20.

The points A and B have position vectors $(3\mathbf{i} + 2\mathbf{j})$ metres and $(6\mathbf{i} - 4\mathbf{j})$ metres respectively. The vectors $\mathbf{i}$ and $\mathbf{j}$ are in a horizontal plane.

- (a) A particle moves from A to B with constant velocity $(\mathbf{i} - 2\mathbf{j}) \text{ m s}^{-1}$. Calculate the time that the particle takes to move from A to B .

(3)

- (b) The particle then moves from B to a point C with a constant acceleration of $2\mathbf{j} \text{ m s}^{-2}$. It takes 4 seconds to move from B to C .

- (i) Find the position vector of C .

(4)

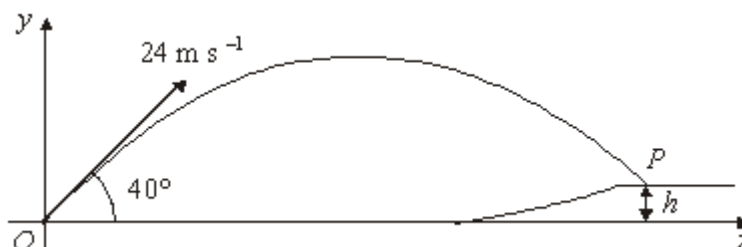
- (ii) Find the distance AC .

(2)

(Total 9 marks)

Q21.

A golf ball is struck from a point O with velocity 24 m s^{-1} at an angle of 40° to the horizontal. The ball first hits the ground at a point P , which is at a height h metres above the level of O .



The horizontal distance between O and P is 57 metres.

- (a) Show that the time that the ball takes to travel from O to P is 3.10 seconds, correct to three significant figures.

(3)

- (b) Find the value of h .

(3)

- (c) (i) Find the speed with which the ball hits the ground at P .

(5)

- (ii) Find the angle between the direction of motion and the horizontal as the ball hits the ground at P .

(2)

(Total 13 marks)

Q22.

A stone is dropped from a high bridge and falls vertically.

- (a) Find the distance that the stone falls during the first 4 seconds of its motion.

(3)

- (b) Find the average speed of the stone during the first 4 seconds of its motion.

(2)

- (c) State one modelling assumption that you have made about the forces acting on the stone during the motion.

(1)

(Total 6 marks)

Q23.

A car travels along a straight horizontal road. The motion of the car can be modelled as

three separate stages.

During the first stage, the car accelerates uniformly from rest to a velocity of 10 m s^{-1} in 6 seconds.

During the second stage, the car travels with a constant velocity of 10 m s^{-1} for a further 4 seconds.

During the third stage of the motion, the car travels with a uniform retardation of magnitude 0.8 ms^{-2} until it comes to rest.

- (a) Show that the time taken for the **third** stage of the motion is 12.5 seconds. (2)
 - (b) Sketch a velocity–time graph for the car during the three stages of the motion. (4)
 - (c) Find the total distance travelled by the car during the motion. (3)
 - (d) State one criticism of the model of the motion. (1)
- (Total 10 marks)

Q24.

A particle moves in a horizontal plane, in which the unit vectors **i** and **j** are directed east and north respectively. At time t seconds, its position vector, **r** metres, is given by

$$\mathbf{r} = (2t^3 - t^2 + 6)\mathbf{i} + (8 - 4t^3 + t)\mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t . (3)
 - (b) (i) Find the velocity of the particle when $t = \frac{1}{3}$. (2)
 - (ii) State the direction in which the particle is travelling at this time. (1)
 - (c) Find the acceleration of the particle when $t = 4$. (3)
 - (d) The mass of the particle is 6 kg. Find the magnitude of the resultant force on the particle when $t = 4$. (3)
- (Total 12 marks)

Q25.

A train travels along a straight horizontal track. It is travelling at a speed of 12 m s^{-1} when

it begins to accelerate uniformly. It reaches a speed of 40 m s^{-1} after accelerating for 100 seconds.

- (a) (i) Show that the acceleration of the train is 0.28 m s^{-2} . (2)
- (ii) Find the distance that the train travelled in the 100 seconds. (2)
- (b) The mass of the train is 200 tonnes and a resistance force of 40 000 N acts on the train. Find the magnitude of the driving force produced by the engine that acts on the train as it accelerates. (3)
- (Total 7 marks)**

Q26.

A golf ball is struck so that its initial velocity is 30 m s^{-1} at an angle of 60° above the horizontal. Assume that the ground is horizontal.

- (a) (i) Show that the time of flight of the particle is 5.30 seconds, correct to three significant figures. (3)
- (ii) Find the distance of the ball from its initial position when it hits the ground for the first time. (2)
- (b) In fact, the ball hits a tree at a point which is at a height of 5 metres above the ground. Given that the ball is descending when it hits the tree, calculate the distance of the tree from the point where the ball was struck. (6)
- (Total 11 marks)**

Q27.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. A yacht moves with a constant acceleration. At time t seconds the position vector of the yacht is $\mathbf{r}$ metres. When $t = 0$ the velocity of the yacht is $(2\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$, and when $t = 10$ the velocity of the yacht is $(-\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$.

- (a) Find the acceleration of the yacht. (3)
- (b) When $t = 0$ the yacht is 20 metres due east of the origin. Find an expression for $\mathbf{r}$ in terms of t . (3)
- (c) (i) Show that when $t = 20$ the yacht is due north of the origin. (2)
- (ii) Find the speed of the yacht when $t = 20$.

(4)
(Total 12 marks)

Q28.

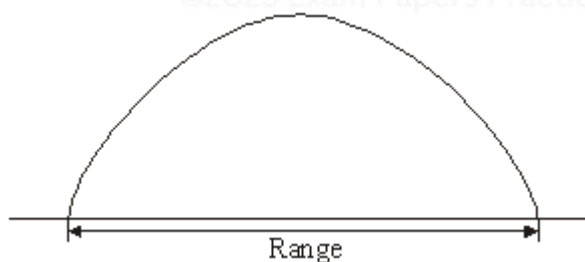
A football is placed on a horizontal surface. It is then kicked, so that it has an initial velocity of 12 m s^{-1} at an angle of 40° above the horizontal.

- (a) State two modelling assumptions that it would be appropriate to make when considering the motion of the football. (2)
- (b) (i) Find the time that it takes for the ball to reach its maximum height. (4)
- (ii) Hence show that the maximum height of the ball is 3.04 metres, correct to three significant figures. (3)
- (c) After the ball has reached its maximum height, it hits the bar of a goal at a height of 2.44 metres. Find the horizontal distance of the goal from the point where the ball was kicked. (7)
- (Total 16 marks)

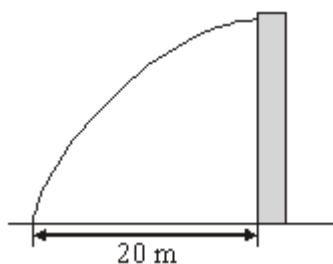
Q29.

A ball is hit from horizontal ground with velocity $(10\mathbf{i} + 24.5\mathbf{j}) \text{ m s}^{-1}$ where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertically upwards respectively.

- (a) State two assumptions that you should make about the ball in order to make predictions about its motion. (2)
- (b) The path of the ball is shown in the diagram.



- (i) Show that the time of flight of the ball is 5 seconds. (4)
- (ii) Find the range of the ball. (2)
- (c) In fact the ball hits a vertical wall that is 20 metres from the initial position of the ball.



Find the height of the ball when it hits the wall.

(4)

- (d) If a heavier ball were projected in the same way, would your answers to part (b) of this question change? Explain why.

(2)

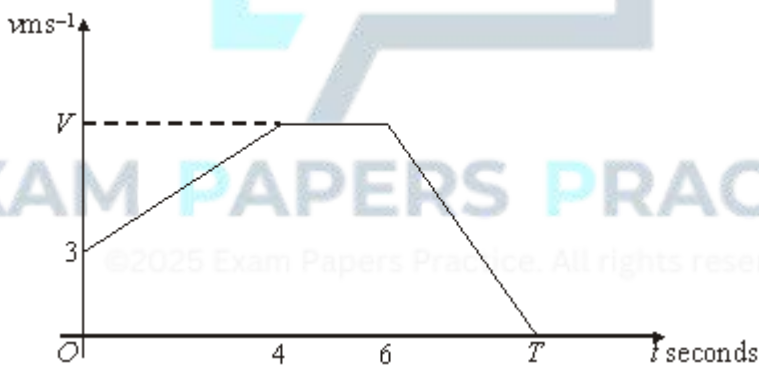
(Total 14 marks)

Q30.

The velocity–time graph below models the motion of an athlete, Paula, during part of a training exercise.

When $t = 0$ she has velocity 3 m s^{-1} . She accelerates at a constant rate until $t = 4$ and her velocity is $V \text{ m s}^{-1}$.

She runs with velocity $V \text{ m s}^{-1}$ for 2 seconds and then decelerates at a constant rate of $f \text{ m s}^{-2}$ until she comes to rest when $t = T$.



- (a) Between $t = 0$ and $t = 4$, Paula runs a distance of 20 metres. Show that $V = 7$.

(2)

- (b) The total distance Paula runs between $t = 4$ and $t = T$ is 35 metres.

- (i) Find the value of T .

(4)

- (ii) Find the value of f .

(2)

- (c) Find Paula's average speed between $t = 0$ and $t = T$.

(2)
(Total 10 marks)

Q31.

A car moves in a straight line up a hill which is inclined at an angle of α to the horizontal, where $\sin \alpha = \frac{1}{14}$. The mass of the car is 1200 kg.

- (a) During the first part of its motion, the car moves with constant speed and is subject to a propulsive force of 1500 newtons and a resistance force of R newtons.

(i) Draw a diagram showing the forces acting on the car.

(2)

(ii) Show that $R = 660$.

(4)

(iii) The car travels 60 metres in 4 seconds. Write down the speed of the car.

(1)

- (b) During the second part of its motion, the propulsive force ceases to act and the resistance force is still 660 newtons. The car continues to move up the hill until it comes to rest.

(i) Find the magnitude of the retardation of the car in this stage.

(2)

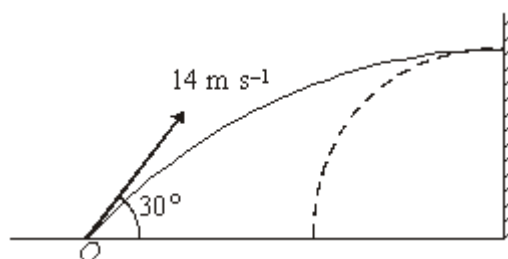
(ii) Find the time between the propulsive force ceasing to act and the car coming to rest.

(2)

(Total 11 marks)

Q32.

Adam kicks his ball from a point O on a horizontal surface towards a vertical wall. He kicks the ball with speed 14 m s^{-1} at an angle of 30° to the horizontal, as shown in the diagram.



When the ball hits the wall, it is travelling **horizontally**.

- (a) Find the vertical height above O at which the ball hits the wall.

(4)

- (b) The ball rebounds horizontally from the wall. Find the time that the ball takes to reach the horizontal surface.

(3)

(Total 7 marks)

Q33.

The velocity-time graph below models the first 8 seconds of a journey of a bus.

During the first 3 seconds of its journey, the bus travels from rest with constant acceleration until it reaches a speed of $V \text{ m s}^{-1}$.

During the next 5 seconds of its journey, the bus travels with constant speed $V \text{ m s}^{-1}$.



- (a) (i) During the first 3 seconds the bus accelerates at 0.8 m s^{-2} .

Show that $V = 2.4$.

(2)

- (ii) Find the total distance the bus travels during the first 8 seconds of its journey.

(3)

- (b) After the first 8 seconds of its journey, the bus decelerates at a constant rate until it stops. During this stage of the journey the bus travels a further 4.8 metres.

Find the average speed of the bus during the whole journey.

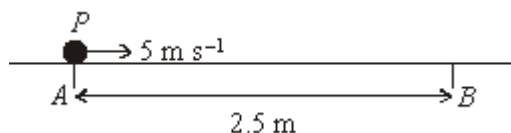
(4)

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(Total 9 marks)

Q34.

A particle P moves in a straight line across a horizontal surface with retardation of magnitude 1.8 m s^{-2} . The particle P passes through a point with velocity 5 m s^{-1} , as shown in the diagram. It subsequently passes **twice** through the point B , where $AB = 2.5$ metres.



- (a) Find the velocities of P on the two occasions when it passes through B .

(3)

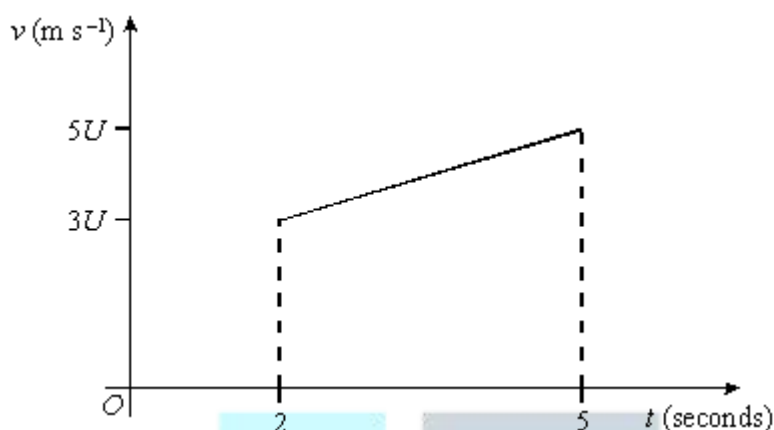
- (b) Hence, or otherwise, find the length of time between the two occasions when P passes through B .

(3)

(Total 6 marks)

Q35.

The velocity–time graph shows the motion of a particle P moving with constant acceleration.



At times $t = 2$ and $t = 5$, P has velocities $3U$ and $5U$ respectively.

- (a) (i) Find, in terms of U , the acceleration of P .
- (ii) Find, in terms of U , the distance travelled by P between the times $t = 2$ and $t = 5$.

(2)

(2)

- (b) When $t = 5$, the motion of P changes and subsequently P moves with a constant retardation. The particle P comes to rest after travelling a **further** 20 metres in the next 4 seconds.

Find the value of U .

(3)

(Total 7 marks)